

Project Description:

In this project, you will design and build a robotic system that can perform a task autonomously via the use of kinematics. The system should include a microcontroller. You will program the microcontroller to control the robot's movements and to process sensor data.

- The project must be approved by the course instructor.
- The deadline for the project proposal submission will be the 6th week.
- The deadline for the final project submission will be the 13th week.

Requirements:

- Design and Construction: The project must include a custom design for the robot and must be constructed using materials such as aluminum or plastic. The robot should be able to move on wheels, treads, or legs, and it should be capable of performing the task specified by the project.
- Microcontroller Programming: The project must include programming of a microcontroller to control the robot's movements and to process sensor data. The programming language can be either C/C++ or a language supported by the microcontroller's IDE.
- Task Performance: The robot must be capable of performing the task specified by the project, such as navigating through a maze, collecting and delivering objects, or detecting and avoiding obstacles.
- Presentation and Documentation: The project must include a presentation of the robot's performance and a detailed documentation of the design, construction, and programming process.

Grading Rubric:

Criteria	Points
Design and Construction	30
Microcontroller Programming	30
Task Performance	30
Presentation and Documentation	10
Total	100

Design and Construction (30 points):

- Custom design of the robot that is capable of performing the specified task (15 points)
- Use of materials such as aluminum or plastic that are appropriate for the robot's function (10 points)
- Construction that is stable, durable, and aesthetically pleasing (5 points)

Microcontroller Programming (30 points):

- Use of appropriate programming language and IDE (5 points)
- Control of the robot's movements and processing of sensor data (15 points)
- Well-structured and documented code (10 points)

Task Performance (30 points):

- Ability of the robot to perform the specified task autonomously (20 points)
- Efficiency, accuracy, and consistency in task performance (10 points)

Presentation and Documentation (10 points):

- Clear and concise presentation of the robot's performance (5 points)
- Detailed documentation of the design, construction, and programming process (5 points)

- **Note: Any copied code from the internet or classmates will result in an instant 0**